

Biplob Biswas

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EXPERIENCE

Senior Research Engineer

Dec. 2023 – Present

Fujitsu Research of America, Inc.

Santa Clara, CA

- Built an **end-to-end agentic evaluation pipeline** for the WoW enterprise benchmark (6K+ tables) using the A2A protocol and MCP-ReAct executors. [*vLLM, OpenAI API, A2A*]
- Engineered an agentic business-process analysis tool, **reducing manual effort (FTE) by 15%** and outperforming top-ranked TableBench model in data analysis task. [*Python, LangGraph, Scikit, Streamlit*]
- Designed and trained a sequence generation model for predictive process monitoring. [*Python, PyTorch*]

Machine Learning Intern

Jun. 2021 – Dec. 2023

Emplifi

Columbus, OH

- Developed a knowledge-grounded **response generation framework** for customer care, deployed as an API. [*AWS, Docker, Python, PyTorch, HuggingFace, FastAPI*]
- Employed neural IR to integrate historical data with a fine-tuned GPT-2, improving **informativeness by 12%** and **semantic similarity by 15.3%** over baselines.

Graduate Research Associate

Aug. 2021 – Dec. 2023

The Ohio State University

Columbus, OH

- Designed a **hybrid (sparse + dense) ranking system** for product-QA RAG, improving **MRR@5 by 10.95%** with 30% lower latency and 38% fewer FLOPs than SOTA cross-encoders. ([Code](#), [Data](#)). [*Python, PyTorch, BERT, T5, FAISS*]
- Developed **TransICD**, an explainable model for ICD coding (MIMIC-III), outperforming LEAM benchmarks in **micro-AUC (0.92)** and micro-F1 (0.64). ([Code](#), [Data](#)). [*Python, PyTorch*]
- Built TransDTBA, a transformer-based model for drug-protein binding affinity prediction. ([Code](#), [Data](#)).
- Implemented joint slot-filling/intent detection and an end-to-end NLG system for shared tasks. ([Code](#), [Data](#): ATIS, E2E).

Engineer, Research and Development

Mar. 2016 – Jul. 2018

KONA Software Lab Limited (in collaboration with KONA I, Seoul, Korea)

Dhaka, Bangladesh

- Developed the front-end for [NexusPay](#) (5M+ downloads), integrating REST APIs for NFC, QR-code, and online payment transactions. [*Android, Java, SQLite*]
- Developed a Nexgo POS application to facilitate secure mobile payment transactions. [*C*]

INDEPENDENT PROJECTS

Raven

Conversational Data Assistant

- Built a backend for natural-language querying of heterogeneous data; orchestrated multi-agent workflow with LangGraph and an optional local LLM to ensure privacy, **reducing manual look-ups by 23%**. ([Code](#))
- Engineered RAG over vector database with persistent memory across multi-turn sessions, served via FastAPI and containerized with Docker Compose. [*LangGraph, PostgreSQL, pgvector, Chroma, Docker, FastAPI*]

EDUCATION

The Ohio State University

Aug. 2018 – Dec. 2023

Ph.D. in Computer Science and Engineering

Columbus, OH

- Advisor: [Dr. Rajiv Ramnath](#)
- Research Topics: Natural Language Processing, Text Generation, Information Retrieval
- Dissertation: [Improving Response Automation for Customer Communication through Document Categorization and Information Retrieval](#)
- Courses Taken: Machine Learning, Data Mining, Parallel Computing, Neural Networks, Speech and Language Processing, Social Media and Text Analysis
- GPA: 3.89 / 4.00

The Ohio State University

Aug. 2018 – May. 2023

Master of Science in Computer Science and Engineering

Columbus, OH

- GPA: 3.89 / 4.00

Bangladesh University of Engineering and Technology

Bachelor of Science in Computer Science and Engineering

- GPA: 3.73 / 4.00

Feb. 2011 – Mar. 2016

Dhaka, Bangladesh

PROFESSIONAL SERVICES

- Reviewer: EMNLP 2023 - Workshop BLP, ASONAM 2023
- Led a team of graduate students to employ a **deep reinforcement learning model for a question-answering task** in customer care service.

TECHNICAL SKILLS

Languages: Python, Java, C, C++

Frameworks & Others: PyTorch, LangChain, LangGraph, Streamlit, FastAPI, Scikit, spaCy, MCP, vLLM

Platforms: Android, AWS, Docker, CUDA, Databricks

Tools: Git, Jira, VS Code, Ollama, LangSmith, Cursor, Claude Code

LLMs: BERT, GPT, Gemini, Llama, Qwen, OpenAI APIs

Database: Oracle, MySQL, PostgreSQL, Chroma, Pinecone

PUBLICATIONS

- **Efficient and Interpretable Information Retrieval for Product Question Answering with Heterogeneous Data**, Biplob Biswas, Rajiv Ramnath. In Proceedings of the Seventh Workshop on e-Commerce and NLP @ LREC-COLING, May 2024.
- **Retrieval Based Response Letter Generation For a Customer Care Setting**, Biplob Biswas, Renhao Cui, Rajiv Ramnath. In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL): Human Language Technologies: Industry Track, July 2022.
- **TransICD: Transformer Based Code-wise Attention Model for Explainable ICD Coding**, Biplob Biswas, Hoang Pham, Ping Zhang. In International Conference on Artificial Intelligence in Medicine (AIME) Jun 2021.
- **Towards Simulating Non-lane Based Heterogeneous Road Traffic of Less Developed Countries**, Q. M. Alam, B. Sarker, Biplob Biswas, K. H. Zubaer, T. Toha, N. Reza and ABM A. Al Islam. In ICT4S, May 2018.

HONORS AND AWARDS

- Dean's Merit List in 2nd and 4th year of Bachelors 2013 - 2016
- Education Board Scholarship - for notable performance in SSC and HSC examination 2010

ADDITIONAL INFORMATION

- H-1B visa approved in 2024